

From Coulomb's Law to Electrostatic Potential

Recall the corresponding magnetostatic result written earlier:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{V'} \frac{\mathbf{J}(\mathbf{r}') \times \mathbf{R}_{12}}{R_{12}^3} dv'.$$

Now let us write the analogous expression in electrostatics:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_{V'} \rho(\mathbf{r}') \frac{\mathbf{R}_{12}}{R_{12}^3} dv',$$

where

$$\mathbf{R}_{12} = \mathbf{r} - \mathbf{r}' = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}, \quad R_{12} = |\mathbf{R}_{12}|.$$

We now define the electrostatic potential as

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_{V'} \frac{\rho(\mathbf{r}')}{R_{12}} dv'.$$

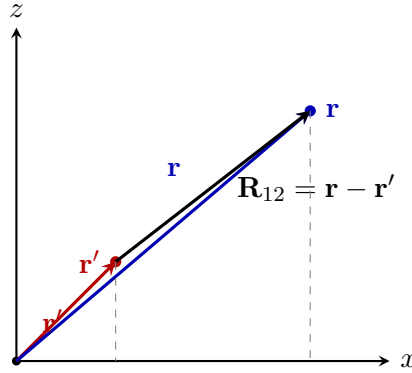


Figure 1: Source point $\mathbf{r}'$ and observation point $\mathbf{r}$. While taking ∇ , the source coordinates (x', y', z') are held fixed, and differentiation is performed with respect to the observation coordinates (x, y, z) .

Also, in electrostatics,

$$\mathbf{E} = -\nabla\phi.$$

Hence,

$$\frac{1}{4\pi\epsilon_0} \iiint_{V'} \rho(\mathbf{r}') \frac{\mathbf{R}_{12}}{R_{12}^3} dv' = -\nabla \left[\frac{1}{4\pi\epsilon_0} \iiint_{V'} \frac{\rho(\mathbf{r}')}{R_{12}} dv' \right].$$

Since the gradient is taken with respect to the observation coordinates,

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z},$$

the quantity $\rho(\mathbf{r}')$ is treated as independent of ∇ . Therefore,

$$\frac{1}{4\pi\epsilon_0} \iiint_{V'} \rho(\mathbf{r}') \frac{\mathbf{R}_{12}}{R_{12}^3} dv' = -\frac{1}{4\pi\epsilon_0} \iiint_{V'} \rho(\mathbf{r}') \nabla \left(\frac{1}{R_{12}} \right) dv'.$$

Bringing all terms to one side,

$$-\frac{1}{4\pi\epsilon_0} \iiint_{V'} \rho(\mathbf{r}') \left[\frac{\mathbf{R}_{12}}{R_{12}^3} + \nabla \left(\frac{1}{R_{12}} \right) \right] dv' = 0.$$

Hence, the required identity is

$$\boxed{\nabla \left(\frac{1}{R_{12}} \right) = -\frac{\mathbf{R}_{12}}{R_{12}^3}} \quad (R_{12} \neq 0).$$

To verify this directly, write

$$R_{12} = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}.$$

Then,

$$\frac{\partial}{\partial x} \left(\frac{1}{R_{12}} \right) = -\frac{x - x'}{R_{12}^3}, \quad \frac{\partial}{\partial y} \left(\frac{1}{R_{12}} \right) = -\frac{y - y'}{R_{12}^3}, \quad \frac{\partial}{\partial z} \left(\frac{1}{R_{12}} \right) = -\frac{z - z'}{R_{12}^3}.$$

Therefore,

$$\nabla \left(\frac{1}{R_{12}} \right) = -\frac{(x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}}{R_{12}^3} = -\frac{\mathbf{R}_{12}}{R_{12}^3}.$$

Important Remark

The identity

$$\nabla \left(\frac{1}{R_{12}} \right) = -\frac{\mathbf{R}_{12}}{R_{12}^3}$$

is the key step that allows us to derive the electrostatic potential formulation directly from Coulomb's law. Thus, for a continuous charge distribution,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_{V'} \frac{\rho(\mathbf{r}')}{R_{12}} dv', \quad \mathbf{E}(\mathbf{r}) = -\nabla\phi(\mathbf{r}).$$

So the electrostatic field is a conservative field and can be obtained from a scalar potential.

Magnetic Vector Potential

Now let us revisit the magnetic field expression:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{V'} \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{R}_{12}}{R_{12}^3} dv',$$

where

$$\mathbf{R}_{12} = \mathbf{r} - \mathbf{r}', \quad R_{12} = |\mathbf{R}_{12}|.$$



Using the identity derived earlier,

$$\nabla \left(\frac{1}{R_{12}} \right) = -\frac{\mathbf{R}_{12}}{R_{12}^3},$$

we can write

$$\mathbf{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \iiint_{V'} \mathbf{J}(\mathbf{r}') \times \nabla \left(\frac{1}{R_{12}} \right) dv'.$$

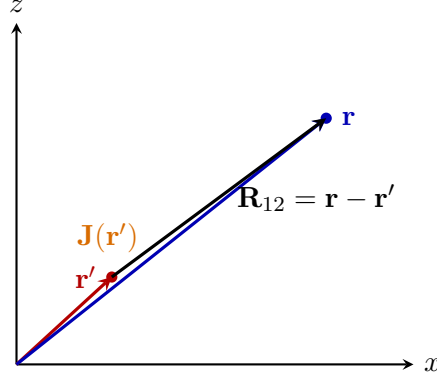


Figure 2: Source current density $\mathbf{J}(\mathbf{r}')$ is defined at the source point $\mathbf{r}'$, whereas the operator ∇ acts with respect to the observation coordinates (x, y, z) .

To proceed further, note that $\mathbf{J}(\mathbf{r}')$ depends only on the source coordinates (x', y', z') , and hence it is treated as a constant with respect to the operator ∇ .

Consider the curl of $\frac{\mathbf{J}(\mathbf{r}')}{R_{12}}$:

$$\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right).$$

Using the vector identity

$$\nabla \times (\phi \mathbf{A}) = (\nabla \phi) \times \mathbf{A} + \phi (\nabla \times \mathbf{A}),$$

with

$$\phi = \frac{1}{R_{12}}, \quad \mathbf{A} = \mathbf{J}(\mathbf{r}'),$$

we get

$$\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right) = \nabla \left(\frac{1}{R_{12}} \right) \times \mathbf{J}(\mathbf{r}') + \frac{1}{R_{12}} \nabla \times \mathbf{J}(\mathbf{r}').$$

But since $\mathbf{J}(\mathbf{r}')$ is independent of (x, y, z) ,

$$\nabla \times \mathbf{J}(\mathbf{r}') = 0.$$

Therefore,

$$\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right) = \nabla \left(\frac{1}{R_{12}} \right) \times \mathbf{J}(\mathbf{r}').$$



Using the anti-commutative property of cross product,

$$\mathbf{J}(\mathbf{r}') \times \nabla \left(\frac{1}{R_{12}} \right) = -\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right).$$

Hence,

$$\boxed{\mathbf{J}(\mathbf{r}') \times \nabla \left(\frac{1}{R_{12}} \right) = -\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right)}$$

Substituting this into the expression for $\mathbf{B}$,

$$\mathbf{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \iiint_{V'} \mathbf{J}(\mathbf{r}') \times \nabla \left(\frac{1}{R_{12}} \right) dv'$$

gives

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{V'} \nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R_{12}} \right) dv'.$$

Since ∇ acts only on the observation coordinates and the integration is over source coordinates, we may bring the curl operator outside the integral:

$$\mathbf{B}(\mathbf{r}) = \nabla \times \left[\frac{\mu_0}{4\pi} \iiint_{V'} \frac{\mathbf{J}(\mathbf{r}')}{R_{12}} dv' \right].$$

This motivates the definition of the **magnetic vector potential**:

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{V'} \frac{\mathbf{J}(\mathbf{r}')}{R_{12}} dv'}$$

Therefore, the magnetic field can be written as

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}}$$

Important Remark

In electrostatics, we introduced a *scalar potential* ϕ such that

$$\mathbf{E} = -\nabla \phi.$$

In magnetostatics, the corresponding potential is a *vector potential* $\mathbf{A}$ such that

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

This difference arises because electrostatic field lines begin and end on charges, whereas magnetic field lines form closed loops.

Final Form

For a continuous current distribution,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{V'} \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dv' \quad \text{and} \quad \mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r}).$$

Divergence of Magnetic Field and Flux Analogy with Electric Field

From the previous result,

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Therefore,

$$\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}).$$

But from vector calculus, we know the identity

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

Hence,

$$\nabla \cdot \mathbf{B} = 0.$$

This is an *always true* result whenever a field can be written as the curl of another vector field.

Important Vector Identity

If a vector field $\mathbf{F}$ is expressible as

$$\mathbf{F} = \nabla \times \mathbf{G},$$

then its divergence is always zero:

$$\nabla \cdot \mathbf{F} = 0.$$

Since $\mathbf{B} = \nabla \times \mathbf{A}$, it follows immediately that

$$\nabla \cdot \mathbf{B} = 0.$$

Integral form through the divergence theorem

Using the divergence theorem,

$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{B}) dv.$$

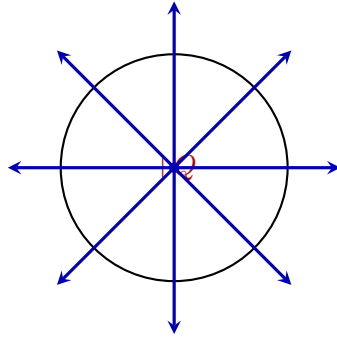
Since $\nabla \cdot \mathbf{B} = 0$,

$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

for *any* closed surface S .

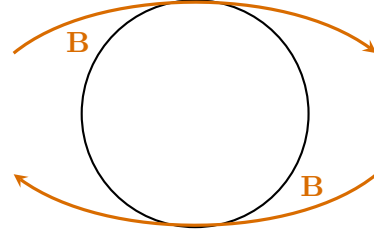
So, the net magnetic flux through a closed surface is always zero.





$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Electric flux can be nonzero



$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Whatever enters must leave

Figure 3: Analogy of flux through a closed surface: electric field can have a nonzero net outward flux if charge is enclosed, whereas magnetic field lines form closed loops, giving zero net magnetic flux through any closed surface.

Physical meaning

The equation

$$\nabla \cdot \mathbf{B} = 0$$

means that magnetic field lines do not start or end inside space. They form continuous closed loops. Therefore, there are no isolated magnetic charges (magnetic monopoles) in classical electromagnetics.

Thus,

No magnetic monopole has been observed in classical electromagnetics.

By contrast, electric field lines originate on positive charges and terminate on negative charges. That is why the divergence of $\mathbf{E}$ is generally not zero.

Flux analogy for $\mathbf{E}$ and $\mathbf{B}$

For electrostatics, Gauss's law gives

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

Applying the divergence theorem,

$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{E} dv = \frac{1}{\epsilon_0} \iiint_V \rho dv.$$

Hence,

$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}$$



where

$$Q_{\text{enc}} = \iiint_V \rho \, dv.$$

For magnetostatics,

$$\nabla \cdot \mathbf{B} = 0.$$

So,

$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{B} \, dv = 0.$$

Hence,

$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = 0.$$

Analogy: Flux Laws

$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad \Longleftrightarrow \quad \oiint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

So, enclosed *charge* produces net electric flux through a closed surface, whereas there is *no net magnetic flux* through any closed surface.

A very important caution

It is tempting to say that “charge is the source of electric field, and current is the source of magnetic field.” This statement is physically useful, but mathematically we must be careful.

- Charge appears in the **divergence** equation of $\mathbf{E}$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

So charge acts as a *source or sink* of electric flux.

- Current appears in the **curl** equation of $\mathbf{B}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (\text{magnetostatics}).$$

So current acts as a *source of circulation* of magnetic field, not of magnetic flux.

Therefore, even though magnetic fields are produced by currents, their divergence is still zero:

Current produces curl of $\mathbf{B}$, not divergence of $\mathbf{B}$.

Integral analogy: flux and circulation

Using Stokes' theorem (will prove in next lecture), the magnetostatic curl equation becomes

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \text{ (Ampere's law: will prove in coming lecture)}$$

where

$$I_{\text{enc}} = \iint \mathbf{J} \cdot d\mathbf{S}.$$

Thus the correct analogy is:

$$\boxed{\iint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}} \quad (\text{charge} \leftrightarrow \text{electric flux})$$

$$\boxed{\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}} \quad (\text{current} \leftrightarrow \text{magnetic circulation})$$

while

$$\boxed{\iint_S \mathbf{B} \cdot d\mathbf{S} = 0.}$$

Final Summary

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \Rightarrow \iint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0 \Rightarrow \iint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \Rightarrow \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

So, **charge is the source of electric flux**, whereas **current is the source of magnetic circulation**. This is why the electric field may have nonzero divergence, but the magnetic field always has zero divergence.